

NETWORK MODELING FOR INFECTIOUS DISEASE SPREAD

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Abstract

The COVID-19 pandemic as well as the high number of new annual cases for infectious diseases such as influenza, show the need for optimization of manufacturing of drugs for infectious diseases to meet market demand. Understanding population behavior through prevalence rate investigation at county level and overtime, will provide an approach to improve on sector anticipation and reaction for manufacturing of drugs for infectious diseases. This project employed Causal Network Modeling (Gaussian Bayesian Network (GBN)), Temporal Network Modeling (Temporal Convolutional Neural Network (TCN)) and Hybrid Network Modeling (HybridTCN-Spatial GBN) in order to understand the social factors that impact the spread of infectious diseases and the nature of the spread across locations and time. With $R^2 = 0.74$, the TCN performs best across the three models with 74% of the variation in the variation in the number of infections explained by the TCN model. Additionally, the most important features for the TCN were lag 3 (infection count from 3 weeks ago) and lag 1 (infection count from 1 weeks ago) with Infection Id (Disease Group), Population Estimate and Income also having notable importance.

I. INTRODUCTION/BACKGROUND

The drug manufacturing sector, similar to other sectors, have different product segments. Chronic vs infectious diseases create product segmentation in which the former are life-long conditions with the latter mostly being fast spreading temporary conditions. Chronic conditions represent the most prevalent medical conditions in the US with up to approximately 76.4% of American adults reporting suffering from 1 or more chronic illnesses [1]. Despite this, infectious diseases have the majority of new annual cases with the flu season in the 2024 to 2025, for instance, having 51 million cases, 710,000 hospitalizations and 45,000 deaths [2]. Manufacturing prioritization, depending on demand, should therefore consider infectious diseases as much as chronic diseases.

II. PROBLEM STATEMENT

Although the drug manufacturing sector has been in existence since the industrial revolution, the COVID-19 pandemic showed a lack in preparedness to respond to market demand for medication for infectious diseases. The sector lacked proper anticipation and reaction to the pandemic. Additionally, the high numbers of annual new cases of infectious diseases such as influenza amplify the inefficiencies in anticipation and reaction to infectious diseases. Understanding population behavior across different locations and overtime represents among the approaches that can improve on the anticipation and reaction to infectious diseases in terms of manufacturing optimization.

A. Data Sources

- 1) CDC-NNDSS API (National Notifiable Diseases Surveillance System) - https://data.cdc.gov/NNDSS/NNDSS-Weekly-Data/x9gk-5huc/data_preview County level data on weekly cases of selected infectious national notifiable diseases. Data Size: 1.83 Million rows, 16 columns. Period: 2022 - 2026. Variables: Reporting Area (states), Current MMWR Year, MMWR WEEK, Label (Infectious

Diseases), Current week, Current week (flag), Previous 52 week Max, Previous 52 weeks Max (flag), Cumulative YTD Current MMWR Year, Cumulative YTD Current MMWR Year (flag), Cumulative YTD Previous MMWR Year Cumulative YTD Previous MMWR Year (flag), LOCATION1, LOCATION2, sort order and geocode.

- 2) Census.gov API - <https://www.census.gov/data/tables/time-series/demo/popest/2020s-counties-total.html> County Level: Data on Population Dynamics relevant for spreading of infections. Data size: 293, 940 rows, 8 columns. Period: 2021 - 2025. Variables: SUMLEV, REGION, DIVISION, STATE, COUNTY, STNAME, CTYNAME, POPULATION ESTIMATES and POPULATION CONCENTRATION ESTIMATES.
- 3) BEA API - <https://bea.gov/data/gdp/gdp-by-county> County Level: Data on Income level for capacity to afford preventions against infections. Data size: 47, 925 rows, 6 columns. Period: 2020 - 2024. Variables: State, County, Year and Income attributes.

III. RELATED WORKS

Product demand in the drug manufacturing industry is more characterized as need given the nature of the market being composed of ailing individuals. The need based nature of demand in the industry is associated with the risk of demand uncertainty [3]. L-shaped decomposition-based algorithms form efficient approaches for optimization given demand uncertainty [3]. Demand for the drug manufacturing industry is often based on the supply chains for health institutions [4]. This perspective exposes drug manufacturers to inefficiencies and poor anticipation as seen during the COVID-19 pandemic, making direct observation of the population important. However, the TOPSIS (Technique for Order of Preference by Similarity of Ideal Solution) approach presents a realistic approach to optimizing drug availability to the market [4]. Drug demand uncertainty and variability represent significant factors in optimization decisions in drug manufacture and availability [5]. The emphasis on importance of demand uncertainty on drug manufacturing optimization also places emphasis on improving demand certainty through direct observation of the population. Event analysis models such as the Discrete Event Simulation (DES) models provide an approach for improvement in the anticipation in the drug manufacturing industry [6]. Hence, having an approach that makes demand predictable would reduce demand uncertainty and improve on the anticipation for the industry to outbreaks of infectious diseases.

IV. METHODOLOGY APPROACH

The project aimed understanding how the spread in the different infections evolves across different locations, overtime and depending on population and income attributes for the given location. Spatial structure modeling gives consideration to where infections happen [7], [8]. The temporal dynamics provides consider to the perspective of when infections happen [9]. The causal relationship modeling considers why infections happen and the network interactions modeling considering how locations influence infections across each other [10]. Consequently, the following models will be employed to derive an understanding of the spread of infectious diseases in the US: Causal Network Modeling (Gaussian Bayesian Network (GBN) with temporal features), Temporal Network Modeling (Temporal Convolutional Neural Network (TCN) with spatial-temporal features) and Hybrid Network Modeling (HybridTCN Spatial GBN, a Spatio-Temporal Causal Model).

V. DATA ANALYSIS

A. Data Collection and Preparation

1) Data Accessing:

- County level data on weekly cases of selected infectious national notifiable diseases data was accessed through CDC-NNDSS API (National Notifiable Diseases Surveillance System) in https://data.cdc.gov/NNDSS/NNDSS-Weekly-Data/x9gk-5huc/data_preview. The resultant data size was 1.83 Million

rows and 16 columns for the period between 2022 and 2026, and the following variables: Reporting Area (states), Current MMWR Year, MMWR WEEK, Label (Infectious 2 Diseases), Current week, Current week (flag), Previous 52 week Max, Previous 52 weeks Max (flag), Cumulative YTD Current MMWR Year, Cumulative YTD Current MMWR Year (flag), Cumulative YTD Previous MMWR Year Cumulative YTD Previous MMWR Year (flag), LOCATION1, LOCATION2, sort order and geocode.

- County level data on population dynamics relevant for spreading of infections was accessed through Census.gov API in <https://www.census.gov/data/tables/time-series/demo/popest/2020s-counties-total.html>. The data size for the resultant data was 293, 940 rows and 8 columns for the period between 2021 and 2025 and the following variables: SUMLEV, REGION, DIVISION, STATE, COUNTY, STNAME, CTYNAME, POPULATION ESTIMATES and POPULATION CONCEN TRATION ES-TIMATES.
- Data on Income level for capacity to afford preventions against infections was accessed through BEA API in <https://bea.gov/data/gdp/gdp-by-county>. Resulting data had 47, 925 rows and 6 columns for the period between 2020 and 2024 with the following variables: State, County, Year and Income attributes.

2) Data Handling:

- For the data from CDC-NNDSS on infections, the data was checked for missing entries with columns with 100% missing entries removed, followed by removal of rows with missing entries. Given the large size of the dataset, removing missing entries does not affect the information in the data. The latitude, longitude, Reporting Area and Location 2 variables were used to create County and State variables (feature engineering) through reversing of geo locator for county mapping of the coordinates. The Previous 52 week Max was also converted to integer type following removal of commas in the entries.
- For the data from Census.gov on population, the county column was first cleaned by removing the “county” string that is added after the actual names for counties in majority of the states. Secondly, the data was converted from wide format to long format to allow for the feature engineering of the year and population estimate variables since each year had a separate column with population estimate for the respective year. State and county columns were then renamed to align with the infections data from CDC-NNDSS.
- In the BEA data, feature engineering was conducted to split the GeoName variable into county variable and state abbreviations with the latter the mapped to the actual state names to create the state variable with the entries set to uppercase.
- The final data handling step was the merging of the three datasets using the year, state and county variables.

3) *Data Preprocessing*: Presented below in Figure 1 is the data preprocessing approach that moved the focus from infectious diseases as a whole, to only the high burden infectious diseases with greater than 50,000 cases over the study period.

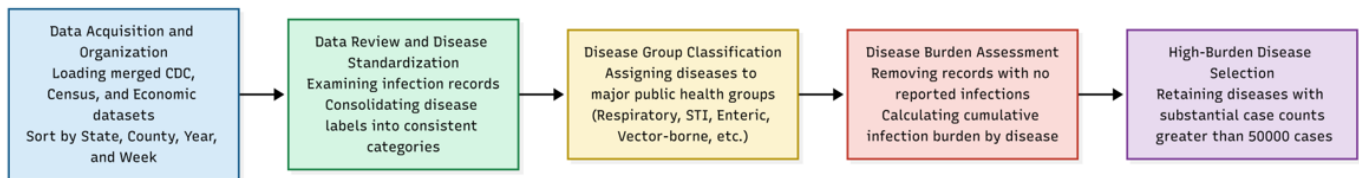


Fig. 1. Data Preprocessing

B. Exploratory Data Analysis (EDA)

The histogram for the distribution of the number of infections for the high burden infectious diseases is presented in Figure 2 below on the left. We note that the data is highly skewed to the right, indicating very

few instances of large outbreaks of infectious diseases at county level. The histogram on right provides a log transformation of the number of infections (used as the target in the modeling); the log transformation improves the distribution by making it less positively skewed.

Distribution of Disease Burden: Raw vs Log Transformation

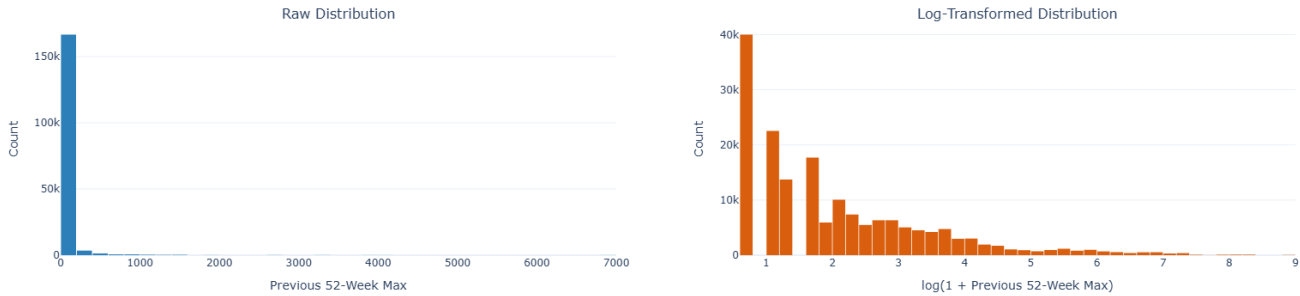


Fig. 2. Distribution of the Number of High Burden Infections

The bar chart below in Figure 3 plots the number of instances counties have recorded occurrence of a high burden infectious disease, representative of outbreak. From the bar chart, Respiratory infections lead in the number of outbreaks across counties followed by Enteric infections (affecting the gastrointestinal tract) while Fungal infections had the least number of outbreaks.

Infections Outbreaks by Disease Group

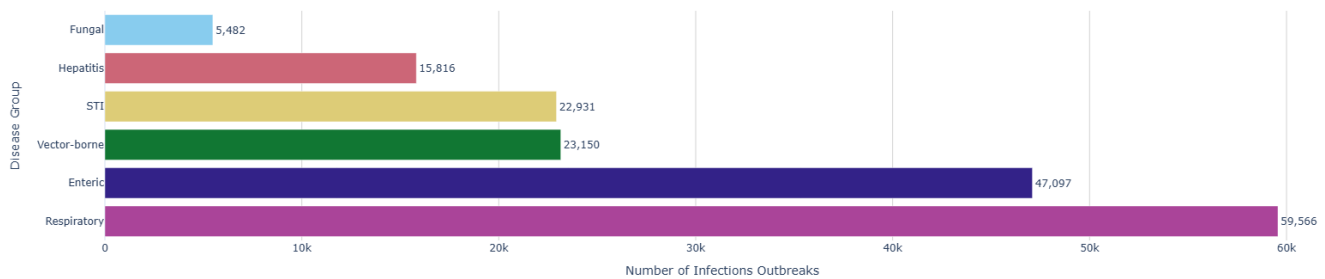


Fig. 3. Distribution of Outbreaks across Infectious Disease Groups

In the bar chart below in Figure 4 for the top counties by total high infectious disease burden, Lake County, Florida in 2024 had the highest number of infections followed by Tulare, California in 2024 as well.

Top Counties by Total Infectious Disease Burden (52-Week Max)

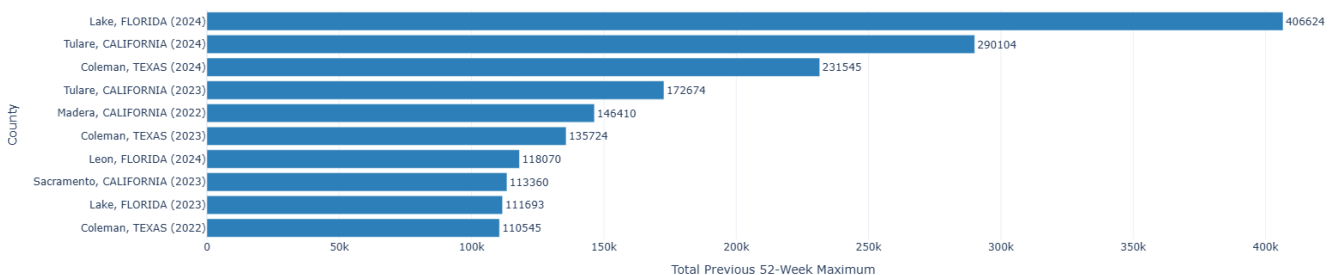


Fig. 4. Top Counties by Total High Burden Infectious Disease

Figure 5 below shows the distribution of high infectious disease burden across weeks and years. Notably, in the 12th Week of the year, there were no recorded high infectious disease burden for 2023. In general, across all weeks, there were higher infections in 2024 as compared to 2022 and 2023.

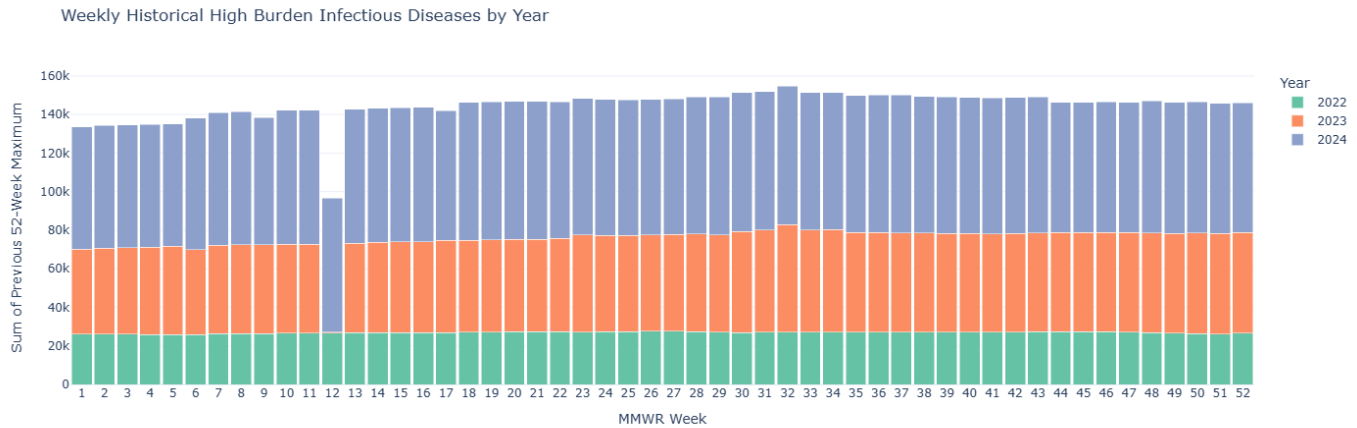


Fig. 5. High Burden Infectious Disease Distribution across Weeks and Years

The distribution of High Infectious Disease Burden across the US shows concentration of outbreaks on the east coast states across the three years, with large outbreaks observed in the west coast states, southern states and east coast states. 2023 is shown to have the highest number of outbreaks as well as the highest number of large outbreaks of high burden infectious diseases.

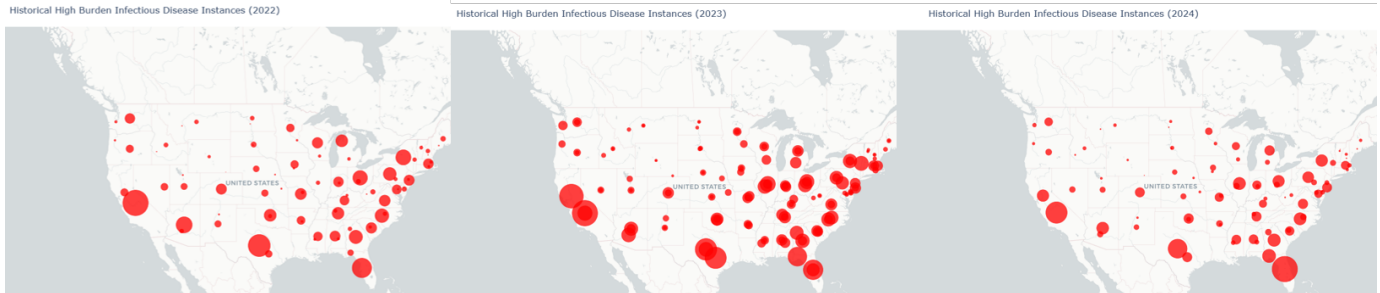


Fig. 6. High Burden Infectious Diseases Mapping

C. Network Modeling

1) Causal Network Modeling: Gaussian Bayesian Networks (GBN): The architecture of the model was such that it had a Bayesian Ridge Regressor and the following parameters: $\alpha_1=1e-6$, $\alpha_2=1e-6$, $\lambda_1=1e-6$, $\lambda_2=1e-6$. $\log(\text{previous 52 week max})$ formed the target with the features being state, county, year, MMWR WEEK, Label, population estimate, income, lag 1, lag 2, lag 3, lag 4. From Table 1, the GBN for the Causal Network Modeling of instances of high burden infectious diseases showed that state, county, year, MMWR WEEK, Label, population estimate, income, lag 1, lag 2, lag 3 and lag 4 accounted for 70.8% ($R^2 = 0.708$) of variation in the instances of infectious diseases. Additionally, Figure 7 shows that the disease group and lag 3 (infection count from 3 weeks ago) were the most important features.

MAE	RMSE	R2
0.587	0.760	0.708

TABLE I
GBN MODEL PERFORMANCE

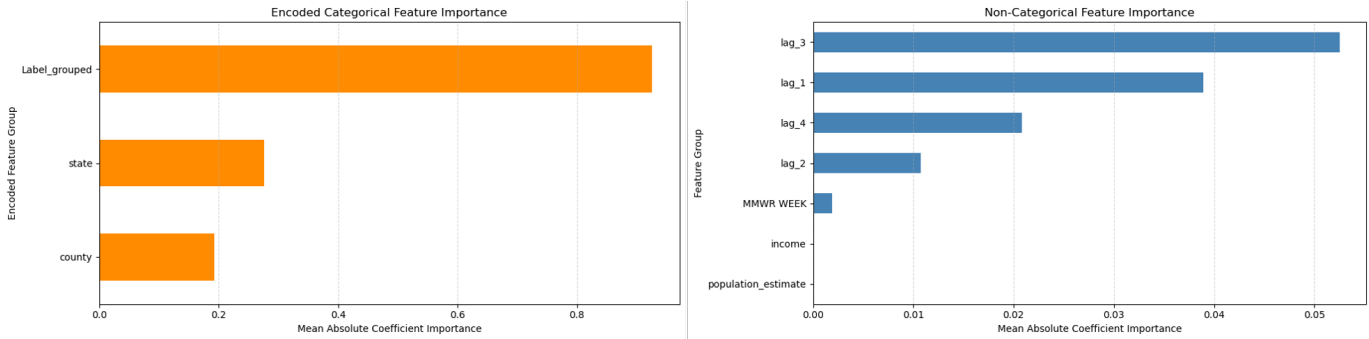


Fig. 7. GBN Feature Importance

2) *Temporal Network Modeling: Temporal Convolutional Neural Network (TCN)*: For the TCN, the model Architecture had Temporal Convolution Neural Network (TCN) with nb filters=64, kernel size=3, dilations=[1, 2, 4, 8, 16, 32] and dropout rate=0.2, and model Learning with 10 epochs and Adam optimizer with lr=0.001. Log(previous 52 week max) formed the target with the following features: State id, county id, year, MMWR WEEK (week sin and week cos), Label (Infection id), population estimate, income, Location (Longitude and Latitude), lag 1, lag 2, lag 3, lag 4. The results in Table 2 indicate that the Temporal Network Modeling (TCN) of instances of infectious diseases showed that State id, county id, year, MMWR WEEK (week sin and week cos), Label (Infection id), population estimate, income, Location (Longitude and Latitude), lag 1, lag 2, lag 3, lag 4 accounted for 73.99% ($R^2 = 0.7399$) of variation in the instances of infectious diseases. Figure 8 showed that lag 3 (infection count from 3 weeks ago) and lag 1 (infection count from 1 weeks ago) being the most important features. Also, from Figure 8, the model learning was characterized by high initial MSE Loss that quickly drops in the first 2 epochs then gentle drops to converge for both the training and validation; with the latter being lower than the former by the same relative margin across all epochs.

MAE	RMSE	R2
0.5479	0.7136	0.7399

TABLE II
TCN MODEL PERFORMANCE

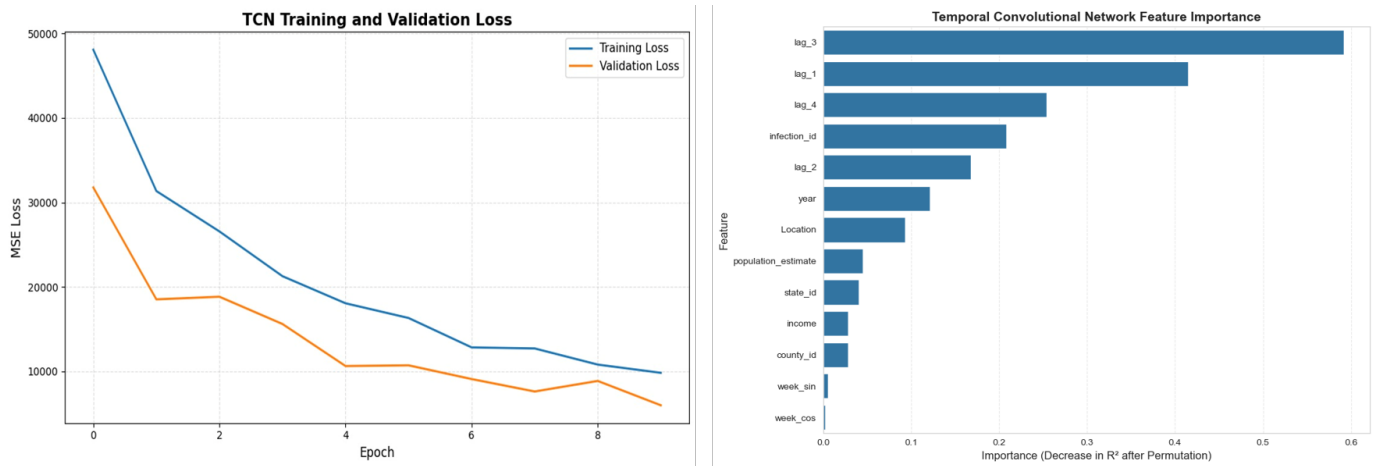


Fig. 8. TCN Feature Importance

3) *Hybrid Network Modeling: HybridTCN Spatial GBN*: The model architecture had:

- Temporal Convolution Neural Network (TCN) with nb filters=64, kernel size=3, dilations=[1, 2, 4, 8, 16, 32], dropout rate=0.2.
- Spatial Encoder Layer with K-Neighbor Spatial Weights with $k = 5$, Inverse distance weighting for neighbor influence, Spatial attention mechanisms
- Gaussian Bayesian Network (GBN)
 - Gaussian mean and variance parameters for each hidden unit
 - Covariance structure learning for variable dependencies
 - Reparameterization trick for differentiable sampling
 - Captures uncertainty in predictions through probabilistic modeling
- Model Learning: Optimizer = Adam with learning rate 1e-3, Loss = Mean Squared Error (MSE), Callbacks (Early Stopping (patience=7), Learning Rate Reduction on Plateau), Batch Size = 256.
- Target: Log(previous 52 week max): log transformed sum of number of infections in past year.
- Features: State id, county id, year, MMWR WEEK (week sin and week cos), Label (Infection id), population estimate, income, Location (Longitude and Latitude), population estimate spatial lag, income spatial lag, lag 1, lag 2, lag 3, lag 4.

The Hybrid Network Modeling (HybridTCN Spatial GBN) of instances of infectious diseases shown in Table 3 below to be such that State id, county id, year, MMWR WEEK (week sin and week cos), Label (Infection id), population estimate, income, Location (Longitude and Latitude), lag 1, lag 2, lag 3, lag 4 accounted for 59.10% ($R^2 = 0.5910$) of variation in the instances of infectious diseases. The actual location and lag 1 (infection count from 1 weeks ago) being the most important features in Figure 9 with the hybrid model shown to under predict the infection instances from the actual and predicted plot.

MAE	RMSE	R2
0.6122	0.8301	0.5910

TABLE III
HYBRIDTCN SPATIAL GBN MODEL PERFORMANCE

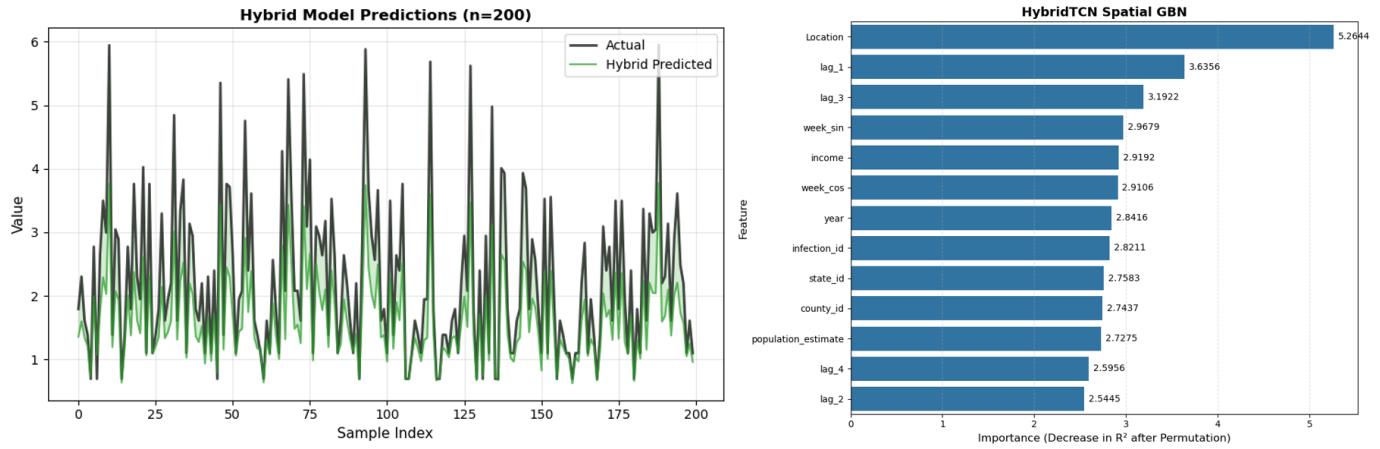


Fig. 9. HybridTCN Spatial GBN Feature Importance

The summary of the model performance across the three models is presented in the grouped chart below. From the results, we note that the TCN model had the highest $R^2 = 0.74$ followed by the GBN model with $R^2 = 0.708$ with the Hybrid model having the lowest $R^2 = 0.591$.

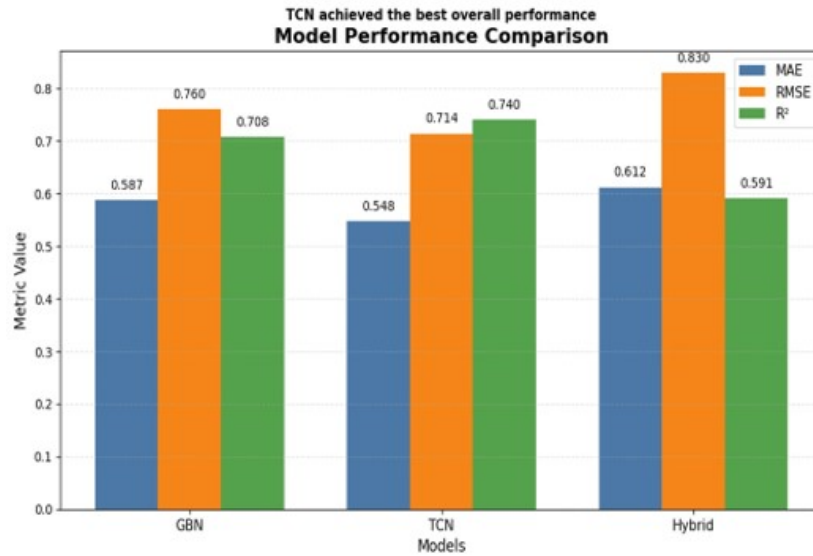


Fig. 10. Model Performance Comparison

VI. CONCLUSION

The Temporal Network Modeling using the TCN of instances of infectious diseases showed best predictive performance with 73.99% ($R^2 = 0.7399$) of variation in the instances of infectious diseases. lag 3 (infection count from 3 weeks ago) and lag 1 (infection count from 1 weeks ago) being the most important features in the Temporal Network Modeling implied that instance of infections mainly depend on outbreaks from 3 weeks ago and 1 week ago. Infection Id (Disease Group), Population Estimate and Income also have notable importance. Infection patterns from 3 weeks ago and from 1 week ago provide information on anticipated infection counts and hence manufacturing of relevant drugs. Additionally, Disease Group, Population Estimate and Income for counties would inform on the manufacturing needs for local markets.

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